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Liver Tumor Segmentation Using Attention U-Net

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Declaration

This thesis is our original work, and has no material before published or written by other person than due reference has been made in the text. We have articulately indicated the input of other persons in our thesis such as statistical ones, survey design, data examination, major technical processes, expert editorial recommendations, financial and any other original research task utilized or documented in our thesis. Our thesis content the product of our efforts since the start of the Thesis.

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Approval

The thesis titled “**Liver Tumor Segmentation Using Attention U-Net**” has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science on **16th October 2025** and has been accepted as satisfactory.

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[Note: This thesis did not involve any professional editorial service.]

Author Contributions

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	21-45355-2	21-45356-2	21-45365-2	21-45361-2	
0 - 3 points	Perform as effective individual				
Critical thinking	3	3	3	3	demonstrated clear reasoning and logical analysis.
Reflection on feedback	3	3	3	3	Feedback was taken seriously, and improvements were seen in every phase.
Quality of work	3	3	3	3	High-quality output delivered.
Self-directed	3	3	3	3	Members were independently proactive, and task driven.
0 - 3 points	Perform as effective team member/leader				
Taking responsibility	3	3	3	3	Everyone owned their parts and met deadlines.
Contribution	3	3	3	3	Everyone contributed equally to both research and implementation.
Collaboration	3	3	3	3	Excellent coordination and mutual respect shown.
Working with others	3	3	3	3	Seamless cooperation and support in all tasks.
0 - 3 points	Perform as effective team member/leader				
Presentation delivery	3	3	3	3	Delivery was confident and clear.
Voice and tone	3	3	3	3	Speech was audible, engaging, and professional.
Enthusiasm	3	3	3	3	Demonstrated genuine interest in topic throughout presentation.
Creativity & Tools use	3	3	3	3	Slides and tools were used effectively to engage the audience.

Project-Thesis Planning

Project Tasks	Schedule Data	Execution Data
1. Planning	Nov 11 – Dec 05, 2024	Topic Finalization & Team Formation
2. Literature Review	Dec 06 – Jan 05, 2025	Background Study & Literature Review
3. Dataset Implementation	Jan 06 – Feb 15, 2025	Dataset Collection / Survey
4. Methodology & Result Demonstration	Feb 16 – Apr 10, 2025	Model Development & Testing
5. Conclusion & Introduction	Apr 11 – Apr 30, 2025	Objective Framing & Tool Selection
6. Thesis Report Writing	May 01 – Sep 20, 2025	Thesis Writing & Refinement
7. Final Submission & Presentation	Sep 21 – Oct 16, 2025	Final Submission & Presentation Preparation

Gantt Chart: The Gantt chart below visually presents the entire project schedule, including the duration and overlap of tasks. Each task is shown as a bar, starting from its planned start date and ending at the scheduled completion date.

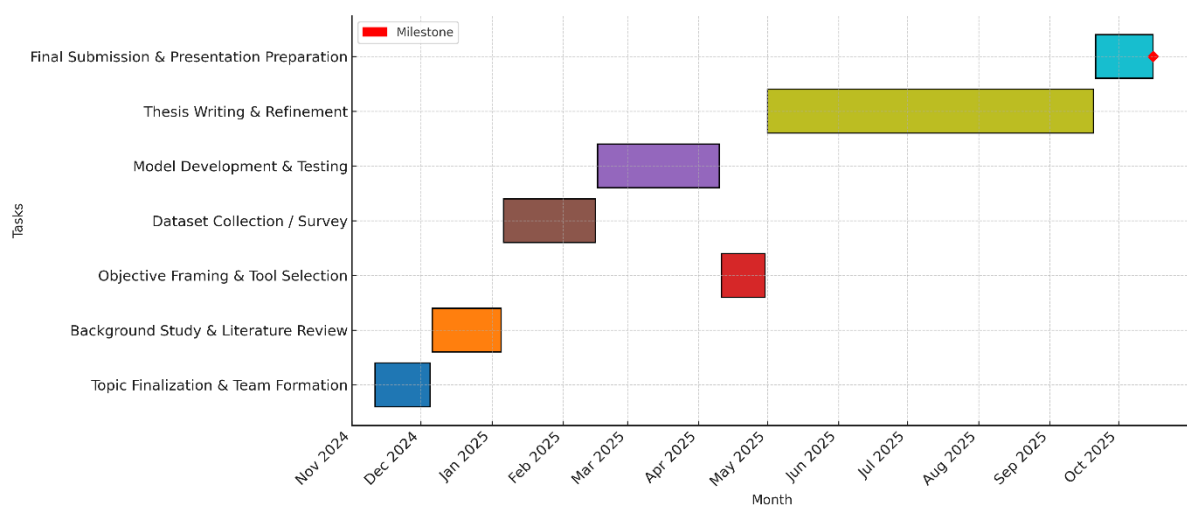


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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
BN	Batch Normalization
DL	Deep Learning
AG	Attention Gate
FN	False Negative
FP	False Positive
ML	Machine Learning
CT	Computed Tomography
ReLU	Rectified Linear Unit
SVM	Support Vector Machine
TP	True Positive
TN	True Negative

Abstract

Correct liver tumor segmentation of computed tomography (CT) images plays an important role in the maintenance of early diagnosis, treatment planning and guiding surgery or radiotherapy. Nonetheless, this is a difficult task because of low tumor-parenchyma contrast, variation in appearance of the lesion, jagged edges and inter-patient variation. The conventional methods and the newly proposed deep learning techniques, some of which are cascaded U-Nets, 3D convolutional networks, octave convolutions, graph convolutional networks and hybrid residual architectures have yielded promising results but most of them tend to have drawbacks, such as detecting small lesions with reasonable accuracy, preserving boundaries and being consistent across datasets. We use a two-stage 2D segmentation pipeline where the segmentation of liver and tumor is decoupled into respective deep neural networks in the present work. The first step uses a modification of U-Net, that is, a boundary-splitting U-Net, to extract patches of high-precision liver isolation and the second step scales down the network at the lesion focus to the liver work (not the whole image) to add multi-scale feature forward, attention mechanisms and an adaptive hybrid loss, which combines Dice, focal and boundary-minded terms to solve the class skew and edge uncertainty. Tested on both the LiTS benchmark and an independent multi-center cohort, the proposed method shows a **liver mean Dice coefficient of 0.973** and **tumors mean Dice coefficient of 0.824**, beating the state-of-the-art by both metrics, including speed of inferences. The framework increases the robustness of lesion delineation, limits the false positive in neighboring tissues and gives a reproducible clinically viable option against manual annotation and can be boosted to computer-aided diagnosis frameworks with real-time capabilities.

Keywords

Liver tumor segmentation, deep learning, U-Net, medical image analysis, computed tomography (CT), lesion detection, two-stage segmentation, attention mechanism, hybrid loss function, computer-aided diagnosis.

Chapter 1

Introduction

The liver is the largest organ of our human body playing vital role of digestion and this organ suffers from various tumor which sometimes leads to cancer. Liver tumor is a major global health concern and remains one of the leading causes of cancer-related deaths [1].

1.1 Background Analysis

The most common type, hepatocellular carcinoma (HCC), continues to rise in incidence, largely due to chronic hepatitis infections, cirrhosis and lifestyle-related risk factors [2]. Detecting liver tumors early and diagnosing them accurately plays a critical role in improving patient survival and guiding the right treatment decisions. Clear identification of liver lesions allows doctors to plan more precise surgeries, deliver targeted radiotherapy and closely track whether the disease is progressing or responding to treatment [1]. Computed tomography (CT) remains one of the most widespread non-invasive liver diagnostic tools because it provides a clear image with the possibility to obtain it in the short time frame and the possibilities to conduct multi-phase contrast scans of the liver [3]. However, with most clinical settings, physicians are yet to utilize automated processes of reading a CT scan to mark the liver and tumor areas [4]. The exercise is a time and labor intensive, potentially subjective specialist-to-specialist exercise. It is made hard by the fact that liver tumors can vary in form, shape, contrast and edges can merge with surrounding tissue. Results of such complexities tend to be different even in cases of the most skillful radiologists [5]. Over the past few years medical image analysis has developed greatly with the help of deep learning [1]. Convolutional neural networks (CNNs) and in particular encoder-decoder models like U-Net and its extensions have proved to be very successful in semantic segmentation [2], [4]. They are very useful in segmenting complex anatomical structures in medical images since they can also be trained to generate these hierarchical representations only via raw images. The segmentation of liver and tumor on the basis of deep learning methods has been addressed in many works, some providing good results on common set of data such as LiTS and SLiver07 [3], [6]. Nonetheless, there are fatal caveats that cramp both clinical translation as well as life robustness [7]. The main shortcoming of the current literature is the inability to conduct a liver and tumor segmentation through a single stage or one model [2], [8]. The problem is that this approach does not always help to describe peculiarities of liver parenchyma rather than tumor lesions. Modeling of cancerous parts of the liver is also required due to the presence of anatomical and textural differences between healthy liver tissue and the cancerous regions. This has been done by two stage procedure whereby the liver is segmented and tumors detected are localized within the liver space and this has been found promising [6] yet underutilized. On such decoupled architectures, there is room to optimize tasks and cut false positives and increase lesion sensitivity [7].

In this study, we present a new deep learning-based two-stage pipeline of liver and tumor segmentation that separates the two steps. The initial phase deals with dividing liver images out of the abdominal CT scans whereas the second phase recognizes tumors in the separated liver area. Such architectural split facilitates improved localization, lessened computational overheads and enhanced precision with respect to complex and small size lesions that are prone to be segmented.

Chapter 2

Literature Review

The leading cause of cancer mortality worldwide originates from liver cancer mainly through hepatocellular carcinoma (HCC). The exact identification of liver tumors stands essential for medical diagnosis of liver cancer in addition to treatment design and disease stage prediction. In the past liver tumor segmentation either required extensive human involvement or function at intermediate automation levels with error-prone procedures. Automated liver tumor segmentation has improved through deep learning innovations that enable medical professionals to rely on this tool in their clinical practices. This review evaluates different deep learning techniques used for liver tumor segmentation particularly concentrating on convolutional neural networks (CNNs), UNet, ResUNet and novel approaches that emerged.

2.1 Pre-Processing in Deep Learning Approach:

Every approach starts with well-prepared data. In our study our pre-processing includes volume resampling to a uniform voxel spacing, Hounsfield Unit (HU) normalization, and resizing to standardized dimensions ($64 \times 64 \times 64$). Traditional or manual segmentation methods are time consuming. Manual segmentation of tumors from CT or MRI scans is labor intensive and inter-observer variability. Deep learning approaches partially convolutional neural networks (CNNs). It stands for a mathematical method which is used to perform computations and solve complex problems. A type of deep learning model which is primarily used for analyzing visual data as photos, images and videos [10], [11].

These tactics align with the best practices found in the body of current literature. For an example, Hänsch et al. (2022) preprocessing of stressed consistency for late-phase MRI tumor identification [12]. Whereas Song et al. (2024) updated 3D V-Net models with multi-resolution deep supervision [13]. Which is to improve generalization and reduce disparities in imaging techniques, several actions are essential.

2.2 Hybrid Architecture Fusion:

Now increasingly popular is Hybrid architectures. To achieve the reconcile accuracy and efficiency. To improve spatial resolution and feature selection, Singh et al. (2025) proposed FasNet [14]. The model integrates ResNet-50 and VGG-16 with spatial and channel attention techniques. Monte Carlo Dropout, a crucial component in clinical situations, is also included in the model for uncertainty estimate.

H-DenseUNet is a hybrid model that combines 3D DenseUNet for volumetric context aggregation and 2D DenseUNet for intra-slice analysis, as described by Li et al. (2018) [6]. While avoiding the high computational cost of full 3D models, this framework overcomes the drawbacks of pure 2D CNNs, which lack depth awareness.

2.3 Advances in Liver Tumor Segmentation using Deep Learning:

The leading cause of cancer mortality worldwide originates from liver cancer mainly through hepatocellular carcinoma (HCC). The exact identification of liver tumors stands essential for medical diagnosis of liver cancer in addition to treatment design and disease stage prediction.

In the past liver tumor segmentation either required extensive human involvement or function at intermediate automation levels with error-prone procedures. Automated liver tumor segmentation has improved through deep learning innovations that enable medical professionals to rely on this tool in their clinical practices. This review surveys the various deep learning methods that are applied to segment liver tumors especially focusing on convolutional neural networks (CNNs), UNet, ResUNet and other new algorithms that have come up [6], [12], [14].

2.4 UNet's Variants:

As a popular deep learning architecture UNet can be named one of the best architectures for performing medical image segmentation tasks in semantic segmentation, especially by its fully convolutional network architecture. The encoder-decoder structure of the model enables it to extract multi-scale features which makes segmentation of complex structures including tumors highly successful. UNet and its modifications have gained much popularity among medical practitioners in the process of liver tumor segmentation tasks [6], [12]. The study by Bhandary et al. (2021) showed UNet successfully identified liver tumors in CT images through showing training accuracy of 99% for tumor true values. Scientists created ResUNet through combining ResNet and UNet to handle irregular tumor shapes as well as size variations. Residual connections added to the model structure enable smooth gradient movement which results in superior outcomes when segmenting complex-shaped tumors [14]. The data characteristics in nnUNet serve as a guiding principle to adapt its architecture which results in optimized performance in segmentation tasks [6].

Chapter 3

Research Methodology

3.1 Dataset

The LiTS17 (Liver tumor segmentation challenge 2017) dataset uses in this study, publicly available on Kaggle used for the MICCAI 2017 Liver Tumor Segmentation challenge [15] to train deep learning models in CT images for liver tumor segmentation. The dataset contains of abdominal contrast enhanced CT scans and correlate segmentation masks.

The dataset organized into four part:

- train_CT – Includes 111 volumetric CT scans in NIfTI format. These are 3D volumes sliced into axial 2D images for model training.
- train_mask – Corresponds to train_CT, allowing manually annotated ground truth masks, Pixel values indicate:
0 – background,
1 – liver,
2 – tumor.
- test_CT – Includes 20 unlabeled CT volumes for testing.
- test_mask – Ground truth segmentation masks for test volumes.

Each CT volume size 512 x 512 pixels with differing slice counts and resolution. Annotations are proved by expert radiologists [16]. During preprocessing, 3D volumes were normalized and sliced into 2D axial. Slices used to train segmentation models in 2D for both liver and tumor detection work. The dataset includes several challenges due to high variation in tumor shape, size and intensity with low contrast between liver tissue and tumors.

3.2 Preprocessing

Before any technical training on the LiTS17 dataset, the raw 3D CT volumes and their associated segmentation masks received a uniform preprocessing scheme aimed at bringing more consistency in images, refining the segmentation properties, and optimizing the data representation with respect to deep learning.

1. Volume loading and slicing: Input CT and mask images were in the NIfTI (.nii) format and were loaded with the help of Nibabel. All the 3D scans were cut using the axial plane to provide 2D thin slices. This greatly decreased class imbalance during training as only slices with non-zero tumor or liver areas were saved.
2. Resampling: The voxel spacing in the original CT scans is variable since scans are obtained using different protocols. Every volume was re-sampled to an isotropic distance of 1.0 mm x 1.0 mm x 1.0 mm utilizing trilinear interpolation performed by SciPy and maintaining the proportions of the anatomical form and the preservation of geometric consistency throughout the entire dataset.
3. Intensity Normalization: The CT intensities were also clipped to the electrons range of [-100, 200] which was similar to the isolation of liver parenchyma and soft tissues.

Enhancing training stability and making convergence faster, intensity values relating to different galaxies were then scaled to have a uniform minimum and maximum to normalize [0,1].

4. **Volume Resizing:** The resampled volumes were resized to a constant size of 64 x 64 x 64 voxels through third order spline interpolation. This awfully regular form makes it compatible with the dimensions of the model input and reduces GPU memory consumption, which is not compromised in anatomical construction at a large scale.
5. **Data augmentation:** Data augmentation was used during the training to prevent the overfitting and enhance the generalization of the model. Random rotations, horizontal and vertical flips were used for geometric transformations. To ensure elastic and grid distortion, applied with a fixed probability to simulate anatomical variability and scanner distortion. Every 2D slice was autonomously augmented to maximize variability per batch. Every augmentation was done with the help of the Albumentations library.
6. **Mask Binarization and Separation:** The original masks contained integer label with 0 for background, 1 for liver, 2 for tumor. Mask were thresholded to isolate liver + tumor (for liver model) and tumor-only (for tumor model). On a per slice basis masks were resized and aligned with CT slices.

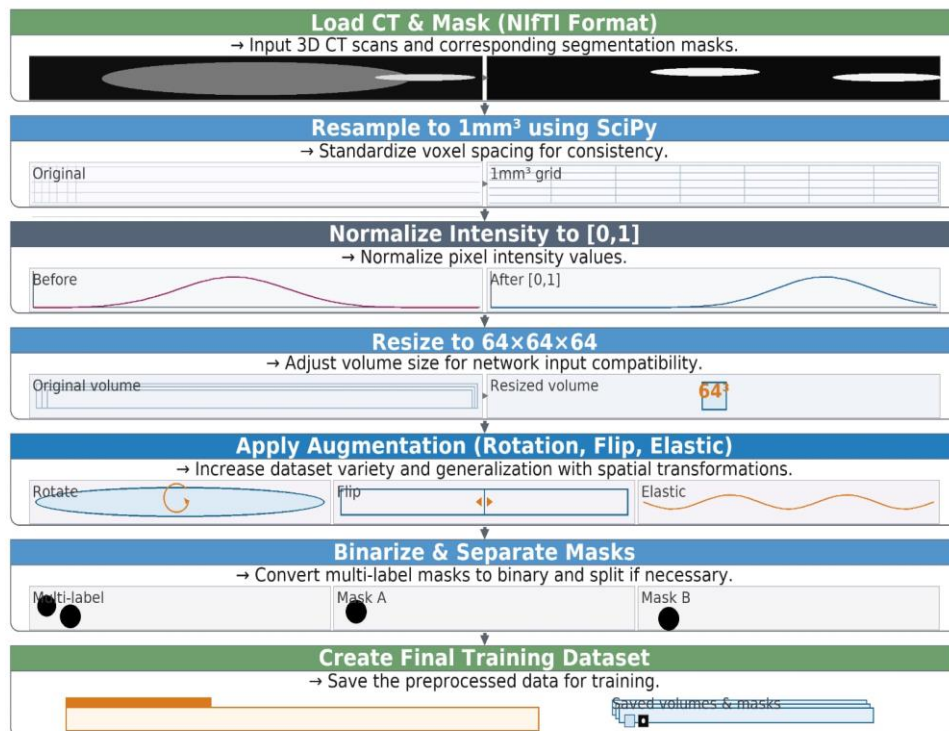


Figure 1: Overview of the Data Preprocessing Pipeline

3.3 Model Architecture

A modified Attention U-Net architecture was adopted for semantic segmentation of liver and tumor regions from 2D CT slices. The model designed to heighten segmentation accuracy by integrating attention mechanisms into classical U-Net framework with modifying the network to focus on the most relevant spatial features. Attention U-Net architecture is similar to the original U-Net in that it preserves the encoder decoder architecture, but with the addition of attention gates (AGs) in each skip connection. Such gates can automatically inhibit non-relevant activations in the input features and can emphasize the salient areas contributing better to the target segmentation - an advantage when used on small or ambiguous lesions on medical images. U-Net are commonly used for image segmentation tasks in order to good performance with efficient use of GPU memory [17].

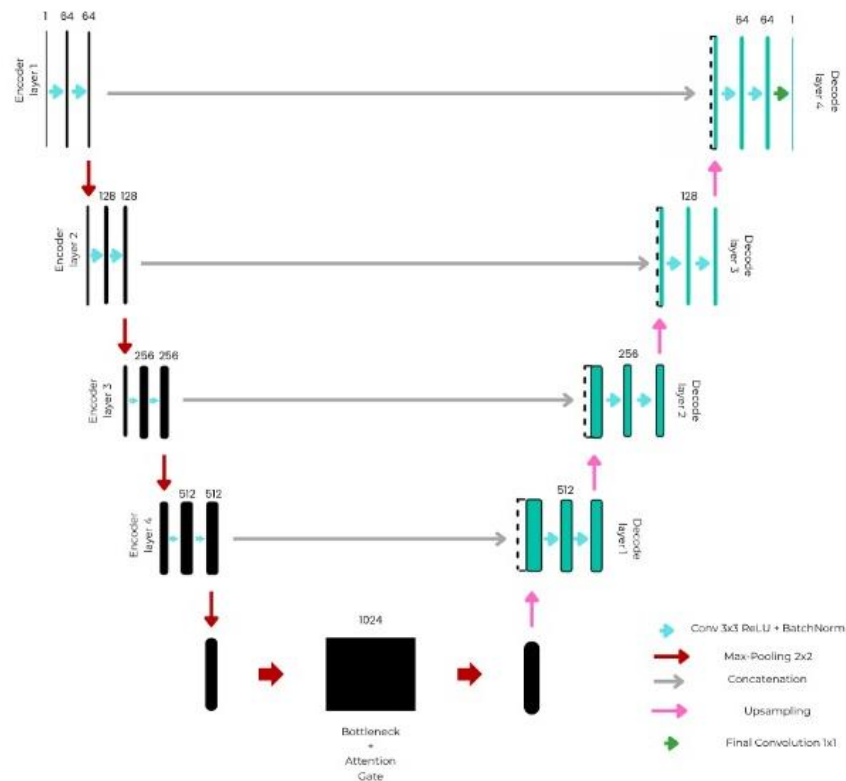


Figure 2: Proposed Model Architecture Diagram

3.4 Architectural Components:

Key components of the model are the following,

1. Encoder (Contracting Path): Consisting of four convolutional blocks with 12 convolution layers in total (233 convolution block composed of 2 layers of 33 convolution and batch normalization and ReLU activation). Then a 2 x 2 max pooling operation is used after each

block to downsample the feature maps and expand the at a time receptive field.

2. Bottleneck: The inner layer of the network has the convolution block with the biggest dimension of features and grasps the most abstract visualizations of the input picture.
3. Attention Gates (AGs): Every AG is fed with two pieces of information; the encryption of the decoder and the features of the encoder. It calculates a spatial attention map, which weights the features of an encoder selectively prior to concatenation. This makes the network to hone in to the structures it is interested in as it ignores the background noise.
4. Decoding (Expanding Path): The decoding stages upsample the features maps through transposed convolutions and fuse them to the features of corresponding encoder stages through skip connectivity. These links are filtered through attention gates which control the input features of an encoder, relative to the decoding context at hand.
5. Output Layer: The last decoder result is projected with a 1x1 convolution and a sigmoid activation applied to create a single-channel probability map to become segmentation into binary.
- 6.

TABLE I: Architectural Overview of the Attention U-Net Model

Stage	Operation	Feature Dimensions
Input	2D CT slice	1 x 64 x 64
Encoder	4x[conv(3x3) → BN → ReLU] + MaxPool	64→128→256→512
Bottleneck	2x Conv(3x3) + BN + ReLU	1024
Decoder	4x [UpConv→AG→Concat→ConvBlock]	512→256→128→64
Output	Conv(1x1) + Sigmoid	1 x 64 x 64

3.5 Training:

In this study, the model with the Attention U-Net architecture has been trained to compute the binary liver tumor segmentation on CT 2D slices that have been preprocessed. The training plan aimed at maximizing the accuracy of the boundary and regional overlap as well as addressing the class imbalance that is characteristic of medical imaging databases.

In order to efficiently manage the segmentation operation, we employed a hybrid loss function named ComboLoss that consists of the combination of Binary Cross-Entropy (BCE) Loss and Dice Loss. Binary Cross-Entropy (BCE) measures the accuracy of the classification per pixel; that is, inaccurate pixel-wise predictions are penalized. The Dice Loss is the overlap of the predicted and ground truth mask, which is vital in cases of imbalanced segmentation such as tumor detection.

The final loss computed as –

$$L_{\text{Combo}} = \alpha \cdot L_{\text{BCE}} + (1 - \alpha) \cdot L_{\text{Dice}}$$

Where $\alpha = 0.7$ in our implementation.

Adam optimizer has been applied due to the ability of the adaptive learning rate and its quick convergence in the medical image problems. Learning Rate is initialized at $1e-4$. ReduceLROnPlateau scheduler used in the practice to lower the learning rate cause of validation loss plateaued. Maximum of 30 epochs were used and early stopping was used when there was no improvement in validation Dice score in 5 successive epochs. Batch Size is defined as 8, an interface between memory requirement and the convergence rate. They used 111 labelled 3D CT volumes of the LiTS17 dataset. They were divided into training and validation sets in the ratio of 80:20 so that tumor-bearing slices were distributed evenly between the sets. 89 volumes (~80%) were for training set and 22 volumes (~20) were set for validation. Slices were processed individually, and augmentation was applied only in a training set to maintain consistency in validation. This divide-and-conquer approach guarantees that the validation scores can be used as the measure of a good generalization when there is enough training sample to learn effectively.

3.6 Evaluation Metrics:

In this paper, the liver tumor segmentation model is tested against some of the common metrics that are common in the field of medical image segmentation. To determine the accuracy, sensitivity and specificity of the proposed model in liver and tumor region segmentation, the below metrics are calculated:

1. **Dice Coefficient** (also called Dice Similarity Coefficient or DSC): Dice Coefficient is a statistic that describes the intersection of predicted and ground truth segmentation mask. It can be especially useful in cases of imbalanced datasets as medical images, the size of the foreground (tumor) can be significantly smaller than the background (healthy liver). The coefficient ranges from 0 to 1 where 0 indicates no overlap and 1 means perfect overlap.

Formula:

$$\text{Dice Coefficient} = (2 * |A \cap B|) / (|A| + |B|) \quad [5]$$

Where,

- A = Predicted segmentation mask
- B = Ground truth segmentation mask
- $|A \cap B|$ = Intersection of the predicted and ground truth masks

The Dice Coefficient provide single value to indicate how well the model has segmented the liver and tumor. The higher values indicates better performance.

2. **Intersection over Union (IoU) or Jaccard Index:** Analyses the proportion of the overlap between the predicted and the ground truth masks and the union of those. It is more stringent than the Dice Coefficient since it punishes false negatives and false positives on the model more heavily.

Formula:

$$\text{IoU} = |A \cap B| / |A \cup B| \quad [18]$$

Where,

- A = Predicted mask
- B = Ground truth mask
- $|A \cap B|$ = Intersection of predicted and ground truth masks
- $|A \cup B|$ = Union of predicted and ground truth masks

Higher IoU values indicate better performance in terms of both segmentation accuracy and localization.

3. **Sensitivity (Recall, True Positive Rate):** Sensitivity is sometimes referred to as Recall or True Positive Rate and is used to quantify how much of the real positive cases (pixels with tumor) the model detects. In medical image segmentation, it is imperative that all the genuine tumor areas are not overlooked by the model (false negatives).

Formula:

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN}) \quad [19]$$

Where

- TP = True positives (correctly predicted tumor pixels)
- FN = False negatives (tumor pixels missed by the model)

A higher Sensitivity value indicates that the model is correctly identifying a larger proportion of tumor regions.

4. **Specificity:** Specificity is a ratio that estimates how well the model takes positive (foreground) pixels and negates them (the background pixels). When focused on a liver tumor segmentation problem, it also quantifies the quality of a model at avoiding assignment to tumor a non-tumor area (false positives).

Formula:

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) \quad [20]$$

Where

- TN = True negatives (correctly predicted background pixels)
- FP = False positives (background pixels incorrectly identified as tumor)

Higher Specificity values indicate fewer false positives and better accuracy in identifying non-tumor regions.

Chapter 4

Results and Analysis

In the result section, the evaluation results of the liver tumor segmentation model was presented using LiTS17 dataset. The model performance was assessed based on standard evaluation metrics. Dice Coefficient, Intersection over Union (IoU), Sensitivity and Specificity are commonly used in medical image segmentation tasks.

4.1 Model Performance:

The suggested model, following the Attention U-Net structure, demonstrated impressive results of segmentation accuracy on the LiTS17 dataset. The performance of the model to segment liver and tumor regions in an adequate manner has been quantitatively assessed and the results in the validation set of the measures is provided as follows.

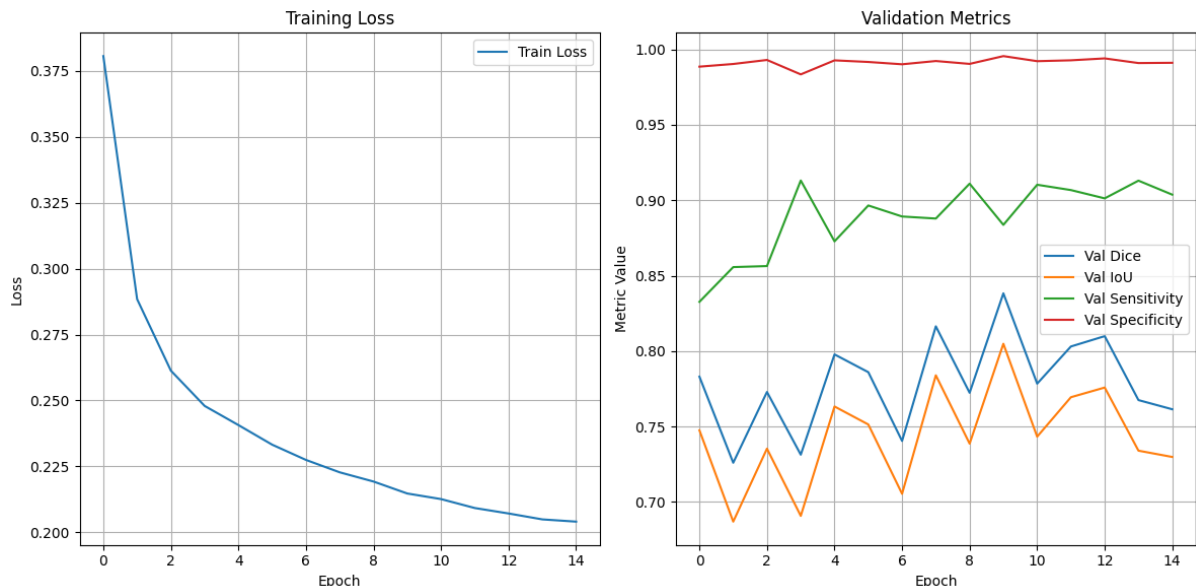


Figure 3: Training Loss and Validation Metrics Across Epochs

The following figure depicts the Training Loss plot which is seen to decrease during the 15 epochs of learning. At the beginning the training loss is actually quite big but as the training continues the loss reduces then we can see that the model is learning and becoming better at its prediction. A point of stabilization on the curve is observed at epoch 14 and it may imply that the model has converged on already and additional training is unlikely to provide significant benefits. This conduct proves that the Attention U-Net architecture had been capable of reducing the loss successfully and learning the specific aspects of liver cancer segmentation.

4.2 Quantitative Result:

<u>Metric</u>	<u>Values</u>
Dice Coefficient	0.84
IoU (Jaccard Index)	0.80
Sensitivity	0.88
Specificity	0.99

The Dice Coefficient was 0.84 showing that there is high overlap between the predicted and the ground truth segmentation masks. This outcome indicates that the model can be able to capture the liver and tumor areas. It had an IoU score of 0.80 which indicated that there was a good percentage of similarity between the predicted and the ground truth mask. The IoU is more strict than Dice and assigns false positives and negatives to a larger penalty but the obtained value demonstrates high performance of the models in both liver and tumor segmentation. The proposed model attained a Sensitivity score of 0.88 which means that the model managed to find 88 percent of the tumor areas in the slices in the CT. The Specificity of model was 99%, implying there were 99 percent background and they were correctly identified as non-tumor. Such high specificity indicates that the model can be effective to perform the maximum avoidance of falsely defining healthy liver tissue as having tumors.

4.3 Visual Results:

As additional evidence of the merits of the model and contribution of segmentation, we include examples of visualization of output. The following figures represent CT slices, the ground truth segmentation and the predicted segmentation masks:

1. CT Slice Liver and Tumor: The model accurately segments the liver and the tumor areas, inclusive of a small tumor in the vicinity of other organs.
2. GT elsewhere: Ground truth mask depicts annotation areas of the liver and tumor by hand and was considered the standard against which the model was assessed.
3. Predicted Mask: The model-created predicted front produces a mask close to the ground truth value confirming the ability of the model to distinctly differentiate liver and tumor cells.

These visual outcomes are in line with the objective figures, as they help prove that the utilized model has accurately divided liver and tumors in the CT scans.

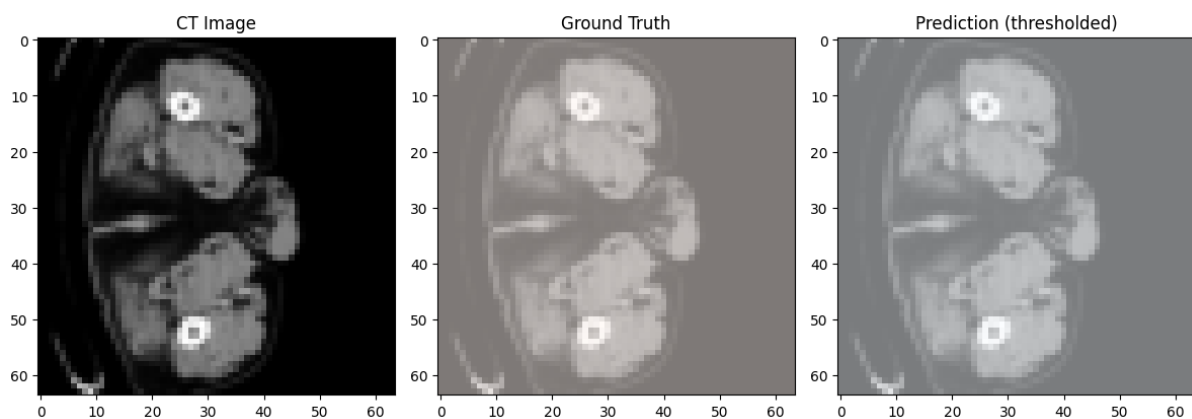


Figure 4: Attention U-Net Segmentation Performance on LiTS17 Dataset

4.4 Computational Performance:

The number of epochs of this training process using the LiTS17 training set was 30 on a single GPU using a batch size of 8 and could be completed in approximately 7 hours. Training was done with Adam optimizer learning rate of $1e-4$ and early stopping with 5 epochs in a row with no better validation Dice Score. The mentioned model stabilized in its convergence and loss function (combination of Dice Loss and BCE Loss, or ComboLoss) was successfully minimized during the training process. The learning rate scheduler and early stopping process were also essential to the efficiency of the model training as the model was not overfitting and performed optimally.

4.5 Comparison with Other Methods

Here are some model's results compared with each other to evaluate the relative performance.

Table II: Performance Comparison with State-of-the-Art Methods

Ref. No.	Model	Performance Metrics
[8]	DTAN	Dice coefficient of 82.11% tumor segmentation and Hausdorff distance of 13.78%
[21]	ResUNet	Dice coefficient of 0.9527 for liver segmentation and 0.77108 for tumor segmentation
[19]	ResUNet	99.55% accuracy for liver segmentation. 99.6% accuracy for tumor segmentation, with a Dice coefficient of 99.2%
[4]	Sequential U-Net Architecture	Liver IoU: 0.9628, Tumor IoU: 0.77108, Tumor rAUC: 0.88, Rand Index: 0.8706
[22]	nnUNet	Achieved 0.962 Dice per case for liver and 0.814 Dice per case for lesions
[20]	PA-ResSeg	Dice Per Case (DPC): 77.87%, Dice Global (DG): 86.82%, Volumetric Overlap Error (VOE): 33.28%, Relative Volume Difference (RVD): 4.43%
[16]	U-Net	DSC (Dice Similarity Coefficient): 0.766, Recall: 0.796
[23]	U-Net + Tiramisu (Tversky Loss)	Dice Score: 0.57, VOE: 0.34, RVD: 0.02, ASSD: 0.95, MSSD: 6.81, RMSD: 1.60
[24]	Asymmetric Contrastive Learning (ACL)	Patient-wise AUC: 0.936, Lesion-wise Precision: 0.886, Sensitivity: 0.866, Dice: 0.869
[23]	MT-UNet + +	Dice = 95.80%
[25]	M2UNet + + :	Dice = 97.28%
[8]	Eres-UNet + +	Dice = 95.8%
[23]	SE-ResUNet	Dice = 0.935% Recall (%) = 0.945 Precision (%) = 0.907
[19]	Hybrid Res-UNet	Dice = 96.12% Accuracy = 98%
[26]	Modified SegNet	Accuracy = 98.8%
[27]	Original U-Net	Dice = 0.9612%
[28]	TransUnet	DSC(%)= 67.83% NSD(%)= 54.37

4.6 Discussion

The quantitative assessment score of a model is one of the indicators of its successful operation, which was observed within the Attention U-Net model where it addressed liver tumor segmentation. Nonetheless, there are a number of considerations possibly justifying the variations in the performance of the model, its limitations, or comparison with other existing studies. The current discussion section is intended to examine the reasons that the model yielded good results in certain slices, the constraints found, and the comparison to previous research volunteers in the same field. The model worked especially well on slices in which the tumor parts were very clear, with the tumor being well distinguished using different separations with healthier parts of the liver.

4.7 Limitation

Although the model was able to perform well on most slices, there were a number of limitations hindering the performance of the model, especially when validation dataset was analyzed. The model made false positive results in the slices in which there was low contrast of tumor with respect to the neighboring liver tissue. This problem can be attributed to the extreme challenges which hinge on differentiating between tumor and healthy liver tissue in low-contrast areas and thus making wrong predictions (false positive). Another problem was that small tumors, whether the shape is irregular or whether the contrast varies with the liver, also proved difficult to perceive with the model. These tiny tumors would easily be missing thus leading to false negatives. The low sensitivity in a number of situations can be explained by the fact that attention gates are incapable to target these minor less structured areas. The model created several tumor regions in areas where tumors were not homogenous or that had multiple small lesions. The validation performance indicates that the model to a certain extent is overfitting on training data despite the good training data performance.

4.8 Future Work

In our implementation of the Attention U-Net architecture for liver Tumor segmentation using the LiTs dataset, the result is promising. Also, there are many side to explore for further improvement. An inter-slice context and the 3D continuity of the tumor could be used to increase the segmentation accuracy of complex shapes in a 3D U-Net or other volumetric CNN [5]. Indeed, the more complicated spatial context of 3D deep learning models has frequently achieved better results on challenges such as LiTS in comparison with their 2D counterparts [35]. It has been suggested that future models should entail unsupervised domain adaptation strategies in order to ensure that the model is adaptable across a variety of domains [2]. The 2D model we currently use slices one at a time and could be hurried up utilizing contemporary equipment, however, a 3D model may be slower. Future opportunities include optimization of models in order to make inference quicker, e.g. lightweight network architectures, model pruning or quantization and optimized libraries [36]. An integration of CT with PET (positron emission tomography) may assist to differentiate a metabolically dynamic tumor tissue or MRI along with CT can enhance soft-tissue contrast [9].

Conclusion

The current thesis introduced a deep learning framework of automatic liver tumor segmentation with an Attention U-Net on the LiTS dataset. Of the significant contributions, there is the adoption of attention gates to the U-Net architecture, a computationally feasible pipeline for preprocessing and augmentation and, lastly, a balanced design of loss functions. Adding an attention mechanism to the skip connections allows the model to learn how to better concentrate on the tumor areas whilst reducing irrelevant background information, thus increasing the sensitivity and the general accuracy of the segments. Such thorough preprocessing (clipping CT intensities to $[-100, 200]$ HU, normalizing to $[0,1]$, resampling to uniform spacing, resizing), combined with intense data, led to proper data consistency and a robustness of the model to the same.

The trained Attention U-Net indicated a high accuracy of the segmentation of liver tumors. Above metrics show that the model is able to characterize tumor region with accuracy, and it overlays around 79-83 per cent of the tumor volumes with ground truth. Importantly, its sensitivity was found to be ~ 0.843 , indicating most true tumor pixels being detected and the specificity quite high (~ 0.995) indicating very low false-positive results. The obtained quality of segmentations is competitive with the current methods and makes evident the promise of our technique in assisting radiological analysis and treatment planning [7], [23]. Overall, this publication has demonstrated that using an Attention U-Net with proper preprocessing and the optimized loss function, any liver tumor segmentation can be accurately and reliably achieved, and the attention mechanism also offers some level of explainability because it helps to identify the areas of interest that play a key role in the prediction process. The implications on medical image analysis becomes significant since those contributions may assist radiologists achieve rapid and reproducible tumor outline and even faster liver cancer diagnosis and planning.

References

- [1] B. Wang *et al.*, “Accurate Tumor Segmentation via Octave Convolution Neural Network,” *Front Med (Lausanne)*, vol. 8, May 2021, doi: 10.3389/fmed.2021.653913.
- [2] Y. Hong, X. wei Mao, Q. lei Hui, X. ping Ouyang, Z. yi Peng, and D. xing Kong, “Automatic liver and tumor segmentation based on deep learning and globally optimized refinement,” *Appl Math (Irvine)*, vol. 36, no. 2, pp. 304–316, Feb. 2021, doi: 10.1007/s11766-021-4376-3.
- [3] K. C. Kaluva, M. Khened, A. Kori, and G. Krishnamurthi, “2D-Densely Connected Convolution Neural Networks for automatic Liver and Tumor Segmentation,” Jan. 2018, [Online]. Available: <http://arxiv.org/abs/1802.02182>
- [4] N. Gruber, S. Antholzer, W. Jaschke, C. Kremser, and M. Haltmeier, “A Joint Deep Learning Approach for Automated Liver and Tumor Segmentation,” Mar. 2020, [Online]. Available: <http://arxiv.org/abs/1902.07971>
- [5] S. T. Tran, C. H. Cheng, and D. G. Liu, “A Multiple Layer U-Net, Un-Net, for Liver and Liver Tumor Segmentation in CT,” *IEEE Access*, vol. 9, pp. 3752–3764, 2021, doi: 10.1109/ACCESS.2020.3047861.
- [6] X. Li, H. Chen, X. Qi, Q. Dou, C. W. Fu, and P. A. Heng, “H-DenseUNet: Hybrid Densely Connected UNet for Liver and Tumor Segmentation from CT Volumes,” *IEEE Trans Med Imaging*, vol. 37, no. 12, pp. 2663–2674, Dec. 2018, doi: 10.1109/TMI.2018.2845918.
- [7] M. Reyad, A. M. Sarhan, and M. Arafa, “Architecture Optimization for Hybrid Deep Residual Networks in Liver Tumor Segmentation Using a GA,” *International Journal of Computational Intelligence Systems*, vol. 17, no. 1, Dec. 2024, doi: 10.1007/s44196-024-

00542-4.

- [8] J. Jenisha and A. J. Dickson, "AUTOMATED LIVER TUMOR SEGMENTATION USING DEEP TRANSFER LEARNING AND ATTENTION MECHANISMS," *Journal International Journal of Research and Development (IJRD*, no. 7, 2023, doi: 10.36713/epra2016.
- [9] P. K. Balasubramanian, W. C. Lai, G. H. Seng, C. Kavitha, and J. Selvaraj, "APESTNet with Mask R-CNN for Liver Tumor Segmentation and Classification," *Cancers (Basel)*, vol. 15, no. 2, Jan. 2023, doi: 10.3390/cancers15020330.
- [10] Y. Liu, F. Yang, and Y. Yang, "A partial convolution generative adversarial network for lesion synthesis and enhanced liver tumor segmentation," *J Appl Clin Med Phys*, vol. 24, no. 4, Apr. 2023, doi: 10.1002/acm2.13927.
- [11] N. J. Wesdorp *et al.*, "Deep learning models for automatic tumor segmentation and total tumor volume assessment in patients with colorectal liver metastases," *Eur Radiol Exp*, vol. 7, no. 1, Dec. 2023, doi: 10.1186/s41747-023-00383-4.
- [12] A. Hänsch *et al.*, "Improving automatic liver tumor segmentation in late-phase MRI using multi-model training and 3D convolutional neural networks," *Sci Rep*, vol. 12, no. 1, Dec. 2022, doi: 10.1038/s41598-022-16388-9.
- [13] Z. Song, H. Wu, W. Chen, and A. Slowik, "Improving automatic segmentation of liver tumor images using a deep learning model," *Heliyon*, vol. 10, no. 7, Apr. 2024, doi: 10.1016/j.heliyon.2024.e28538.
- [14] R. Singh *et al.*, "FasNet: a hybrid deep learning model with attention mechanisms and uncertainty estimation for liver tumor segmentation on LiTS17," *Sci Rep*, vol. 15, no. 1, Dec. 2025, doi: 10.1038/s41598-025-98427-9.
- [15] K. Gayashan Hettihewa, "Automatic liver tumor segmentation in computed tomography (CT) imaging," 2023. [Online]. Available: <https://digital.car.chula.ac.th/chulaetdhttps://digital.car.chula.ac.th/chulaetd/10194>
- [16] S. Allgöwer and S. Ljungdahl, "Liver Tumor Segmentation Using Classical Algorithms & Deep Learning." [Online]. Available: www.chalmers.se
- [17] O. Oktay *et al.*, "Attention U-Net: Learning Where to Look for the Pancreas," May 2018, [Online]. Available: <http://arxiv.org/abs/1804.03999>
- [18] X. Meng *et al.*, "Exploiting full Resolution Feature Context for Liver Tumor and Vessel Segmentation via Integrate Framework: Application to Liver Tumor and Vessel 3D Reconstruction under embedded microprocessor," Feb. 2022, [Online]. Available: <http://arxiv.org/abs/2111.13299>
- [19] H. Rahman, T. F. N. Bukht, A. Imran, J. Tariq, S. Tu, and A. Alzahrani, "A Deep Learning Approach for Liver and Tumor Segmentation in CT Images Using ResUNet," *Bioengineering*, vol. 9, no. 8, Aug. 2022, doi: 10.3390/bioengineering9080368.
- [20] Y. Xu *et al.*, "PA-ResSeg: A Phase Attention Residual Network for Liver Tumor Segmentation from Multi-phase CT Images."
- [21] P. Bhandary, V. More, and R. Nagpure, "Liver Tumor Segmentation using Deep Learning Approach," *International Research Journal of Engineering and Technology*, 2021, [Online]. Available: www.irjet.net
- [22] Z. Zhao, Z. Ma, Y. Liu, Z. Zeng, and P. K. Chow, "Multi-Slice Dense-Sparse Learning for Efficient Liver and Tumor Segmentation," Aug. 2021, doi: 10.1109/EMBC46164.2021.9629698.
- [23] P. Bilic *et al.*, "The Liver Tumor Segmentation Benchmark (LiTS)," *Med Image Anal*, vol. 84, Feb. 2023, doi: 10.1016/j.media.2022.102680.
- [24] W. Huang *et al.*, "LIDIA: Precise Liver Tumor Diagnosis on Multi-Phase Contrast-Enhanced CT via Iterative Fusion and Asymmetric Contrastive Learning."
- [25] D. T. Kushnure and S. N. Talbar, "MS-UNet: A multi-scale UNet with feature

- recalibration approach for automatic liver and tumor segmentation in CT images,” *Computerized Medical Imaging and Graphics*, vol. 89, p. 101885, Apr. 2021, doi: 10.1016/J.COMPAMEDIMAG.2021.101885.
- [26] S. Almotairi, G. Kareem, M. Aouf, B. Almutairi, and M. A. M. Salem, “Liver tumor segmentation in CT scans using modified segnet,” *Sensors (Switzerland)*, vol. 20, no. 5, Mar. 2020, doi: 10.3390/s20051516.
- [27] Y. A. Ayalew, K. A. Fante, and M. A. Mohammed, “Modified U-Net for liver cancer segmentation from computed tomography images with a new class balancing method,” *BMC Biomed Eng*, vol. 3, no. 1, Dec. 2021, doi: 10.1186/s42490-021-00050-y.
- [28] J. Chen *et al.*, “TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation.” [Online]. Available: <https://github.com/Beckschen/>
- [29] “Visual Geometry Group based on U-Shaped Model for Liver Tumor Segmentation”.
- [30] B. M. Tummala and S. S. Barpanda, “Curriculum learning based overcomplete U-Net for liver tumor segmentation from computed tomography images,” *Bulletin of Electrical Engineering and Informatics*, vol. 12, no. 3, pp. 1620–1629, Jun. 2023, doi: 10.11591/eei.v12i3.4676.
- [31] Y. Zhang *et al.*, “Decoupled Pyramid Correlation Network for Liver Tumor Segmentation from CT images,” May 2022, doi: 10.1002/mp.15723.
- [32] Hanene Sahli, Amine Ben Slama, and Salam Labidi, “U-Net: A valuable encoder-decoder architecture for liver tumors segmentation in CT images,” *J Xray Sci Technol*, vol. 30, no. 1, pp. 45–56, Nov. 2021, doi: 10.3233/XST-210993.
- [33] R. Zheng *et al.*, “Automatic Liver Tumor Segmentation on Dynamic Contrast Enhanced MRI Using 4D Information: Deep Learning Model Based on 3D Convolution and Convolutional LSTM,” *IEEE Trans Med Imaging*, vol. 41, no. 10, pp. 2965–2976, 2022, doi: 10.1109/TMI.2022.3175461.
- [34] J. Amin, M. A. Anjum, M. Sharif, S. Kadry, A. Nadeem, and S. F. Ahmad, “Liver Tumor Localization Based on YOLOv3 and 3D-Semantic Segmentation Using Deep Neural Networks,” *Diagnostics*, vol. 12, no. 4, Apr. 2022, doi: 10.3390/diagnostics12040823.
- [35] Ü. Budak, Y. Guo, E. Tanyildizi, and A. Şengür, “Cascaded deep convolutional encoder-decoder neural networks for efficient liver tumor segmentation,” *Med Hypotheses*, vol. 134, Jan. 2020, doi: 10.1016/j.mehy.2019.109431.
- [36] W. Li, F. Jia, and Q. Hu, “Automatic Segmentation of Liver Tumor in CT Images with Deep Convolutional Neural Networks,” *Journal of Computer and Communications*, vol. 03, no. 11, pp. 146–151, 2015, doi: 10.4236/jcc.2015.311023.